Data Visualization

Week 3

Github and assignment housekeeping

- Assignment tips on Canvas

- Clean up Github folders
  - Your project should contain:
  - Some scripts
  - One . Rproj file
  - A datafile
  - ONE readme.md file that tells you how these things go together

- All of your scripts should run from beginning to end with no errors
- Your README file should only contain a log, not actual code. Put your code in your scripts. (See example here: https://github.com/VTQuantMethodsEEB/klangwig/blob/master/README_example.md)

- **Every time you submit an assignment:**
- You will need to do a new **COMMIT** and type something in the box
- Second, you will need to **PUSH** to share any changes to your files with me and send them to GitHub.
- Each week you will need to **PULL** to see my comments on your scripts. Then, commit and push to send me your assignment.

- **Please annotate your code**
- If you have a question about something, you can ask in your code annotation
- #I would like to print the names of the species that are higher than the mean, but I'm not sure how to do that
- I will (try) to help and push back the script to you with how to do this! (Hopefully…)

Preparation

- Install package “ggplot2”
- Install package “gridExtra”

Fun extras:

- Install package “viridis”
- Install package “cowplot”
- Install package “ggforce”
- Install package ”shiny” (maybe we want to do this as an extension?)
- Install package “ggthemes”
  - This is really awesome – use themes based on plots from *The Economist, STATA, Excel 2003, Fivethirtyeight, WSJ, etc.*

Goals

- Participants will learn to think broadly about the effective uses of visualization
- Understand previous research on visualization
- Learn the ggplot2 package and some extensions

What makes figures compelling?

Some history on data visualization

- There is a HUGE field dedicated to the perception of data (e.g. marketing, business, art)
- Scientists rarely take these classes or know anything about it
- There are a few exceptions that have “broken through” into our field

Cleveland & McGill 1984 Jour. Of the American Statistical Association.

- They outline the problem – graphs are presented haphazardly

- What's in it? The paper describes a series of experimental studies to understand how basic visual primitives (e.g. color, length, size) compare in delivering quantitative information.

- Why is it important? It presented a hierarchy for understanding data to best portray quantitative information.

Hierarchy (most to least accurate in perception of quantitative data)

1. Position along a common scale
2. Positions along non-aligned scales
3. Length, direction, angle
4. Area
5. Volume, curvature
6. Shading, color saturation

Examples of the elementary perceptual task

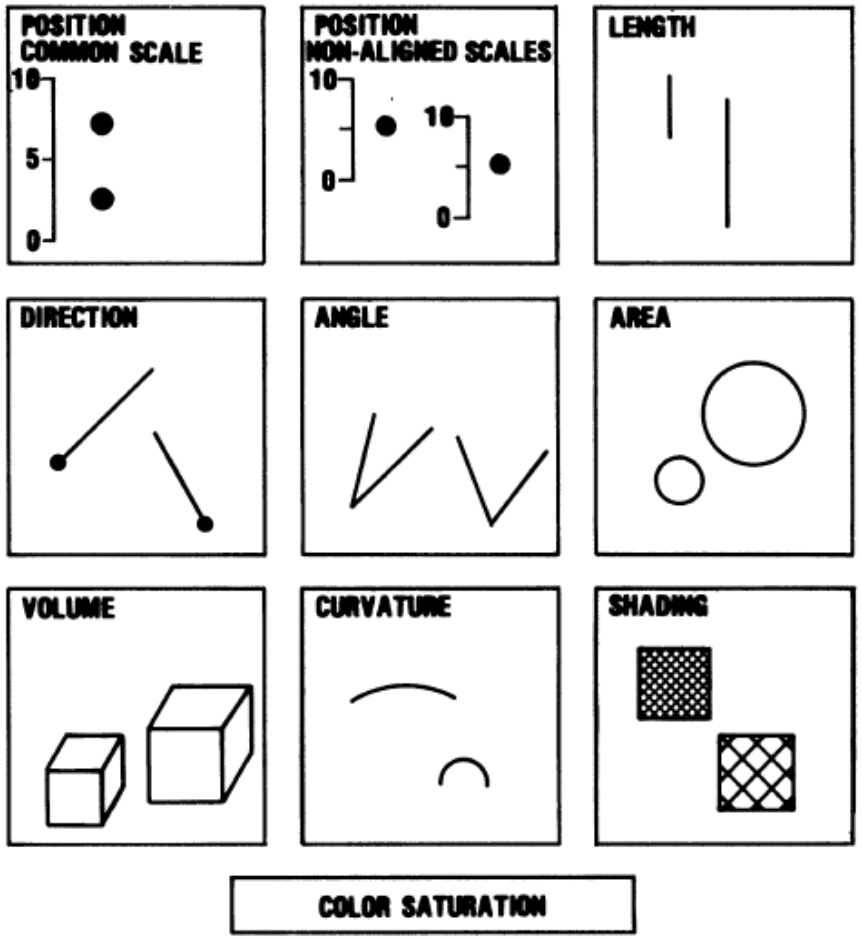


Figure 1. Elementary perceptual tasks.

Saturation: not included because “expense” of color figs

Shading and hue: excluded from study because they do not have an unambiguous method for ordering from large to small

Position-length experiment

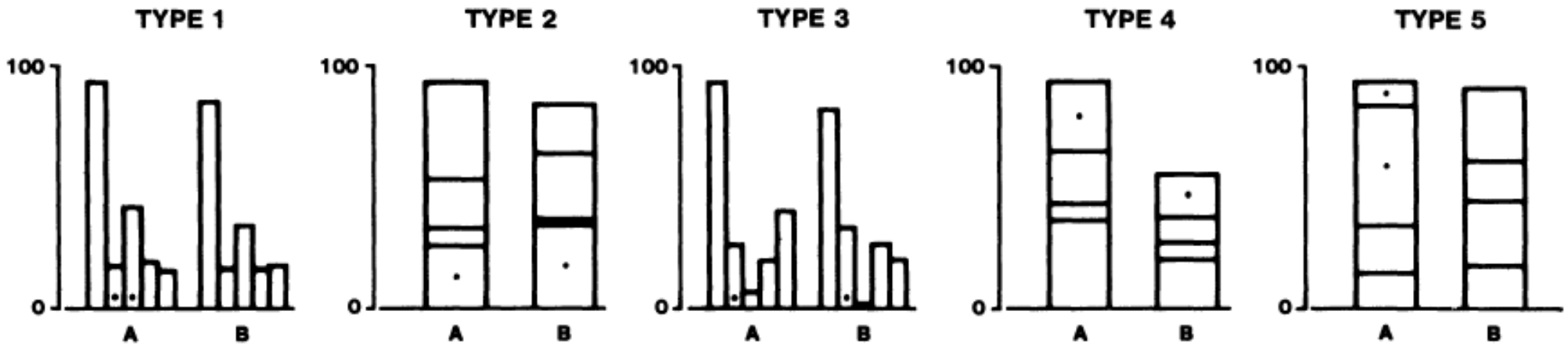


Figure 4. Graphs from position–length experiment.

What percent of the larger is the smaller?

Edward Tufte

Chartjunk: stuff that isn’t necessary to understand the graph

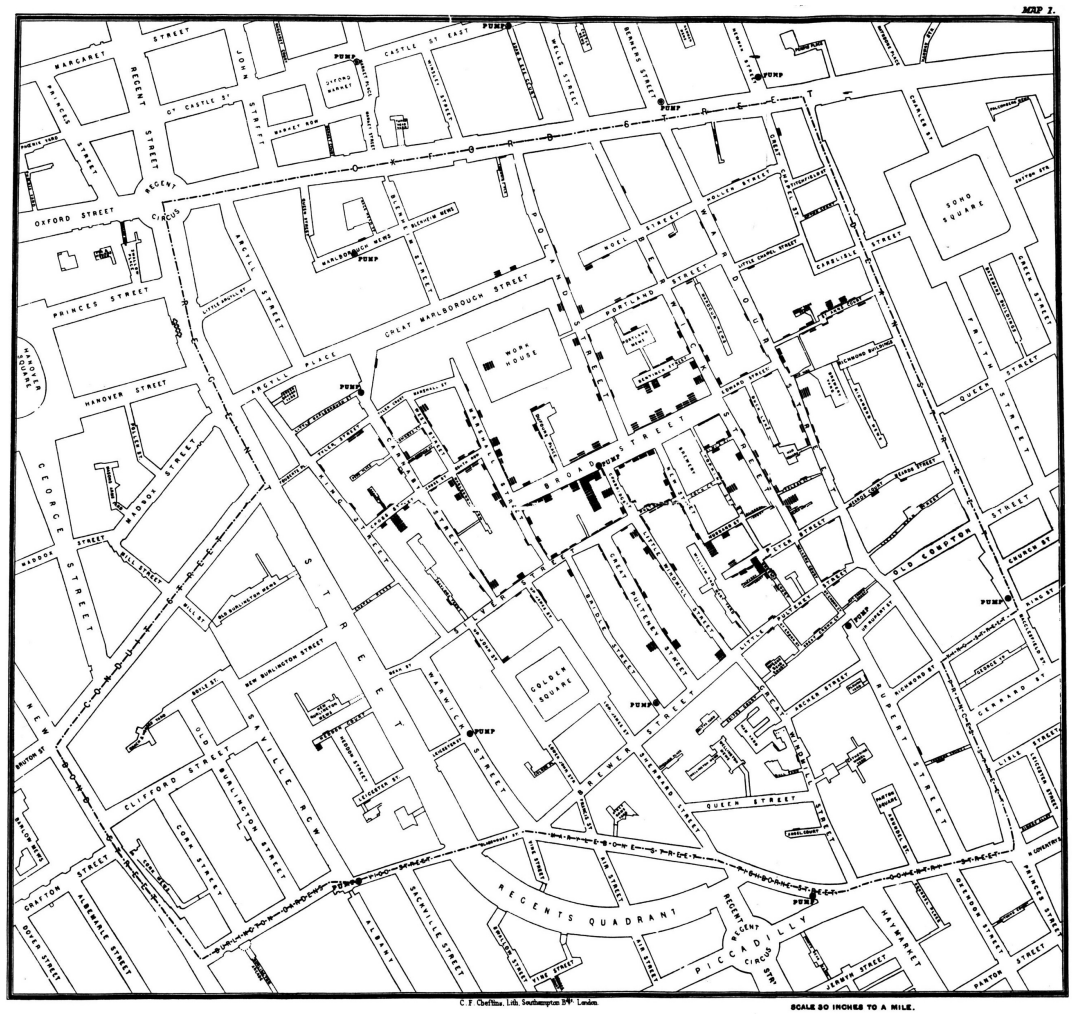
Edward Tufte

- Data-rich visualizations:

 every data point has value, but when observed more generally only trends and patterns can be observed

- High data-ink ratio

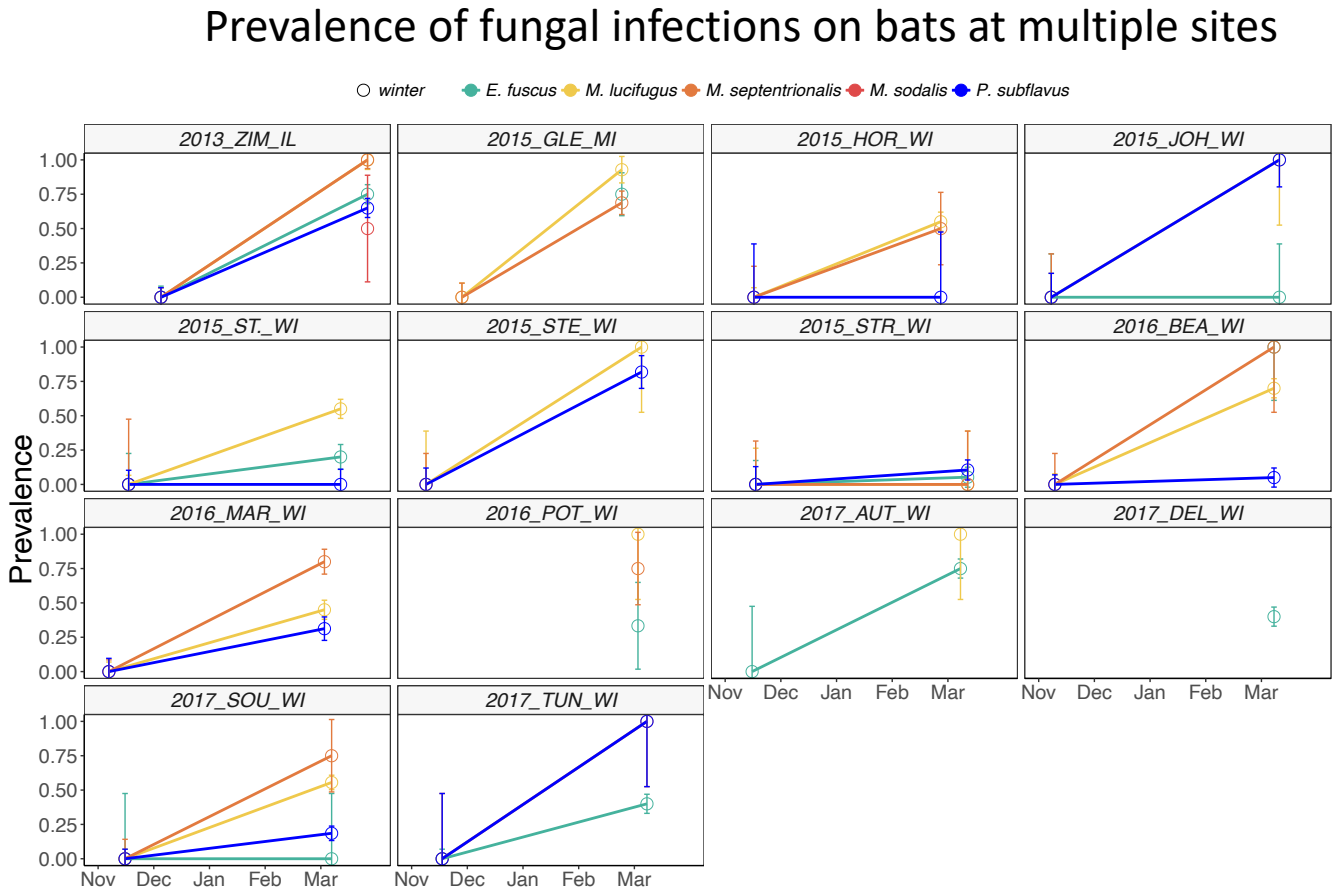
John Snow's map of cholera fatalities



Edward Tufte

- Small multiples:

A series of similar graphs using the same scale or axes



Edward Tufte

- Direct labeling:

Put the text on the figure

Direct labeling

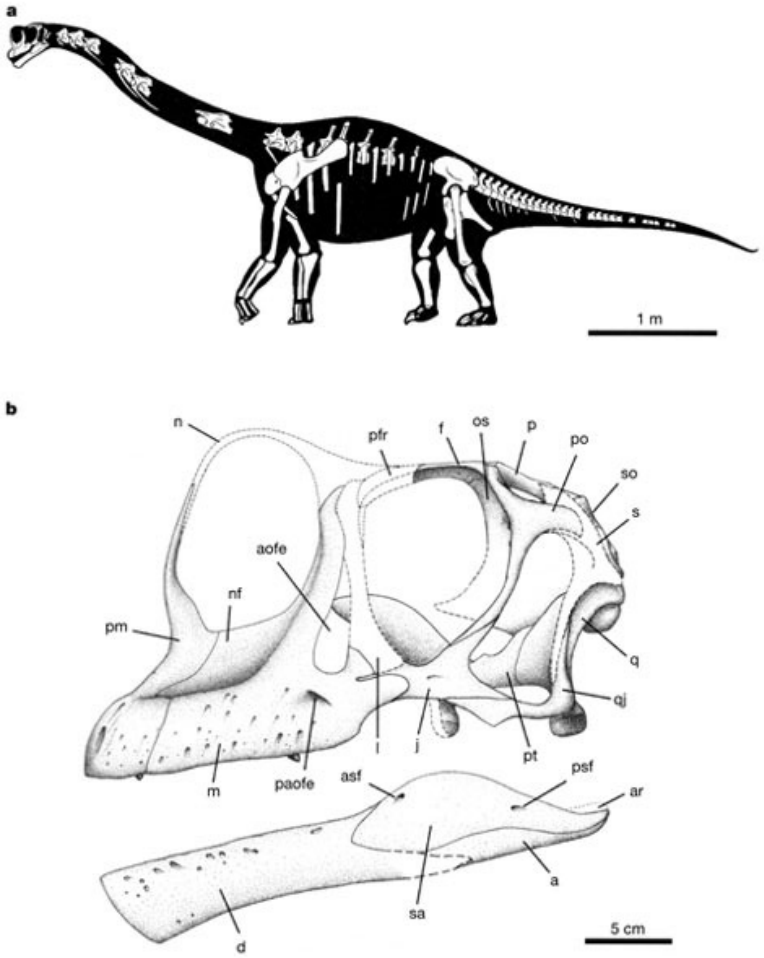
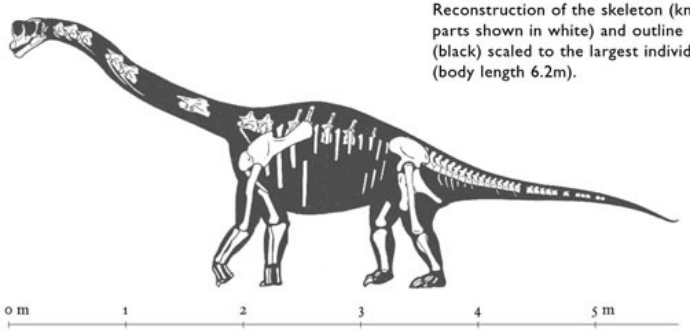
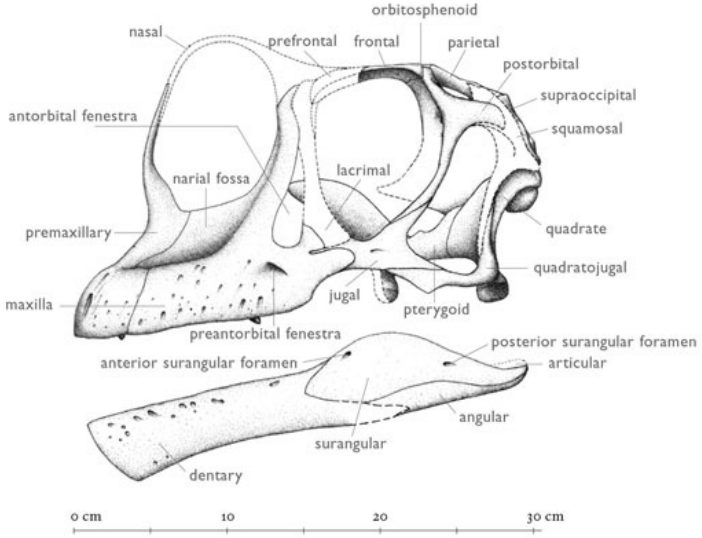


Figure 1 | *Europasaurus holgeri* gen. et sp. nov. a, Reconstruction of the skeleton (known parts shown in white) and outline (black) scaled to the largest individual (body length 6.2 m). **b**, Skull reconstruction based on several individuals but scaled to the holotype. Abbreviations: a, angular; aofe, antorbital fenestra; ar, articular; asf, anterior surangular foramen; d, dentary; f, frontal; j, jugal; l, lacrimal; m, maxilla; n, nasal; nf, narial fossa; os, orbitosphenoid; p, parietal; paofe, preantorbital fenestra; pfr, prefrontal; pm, premaxillary; po, postorbital; psf, posterior surangular foramen; pt, pterygoid; q, quadrate; qj, quadratojugal; s, squamosal; sa, surangular; so, supraoccipital.



***Europasaurus holgeri* gen. et sp. nov.** Reconstruction of the skeleton (known parts shown in white) and outline (black) scaled to the largest individual (body length 6.2m).



Skull reconstruction based on several individuals but scaled to the holotype.

More Tufte info

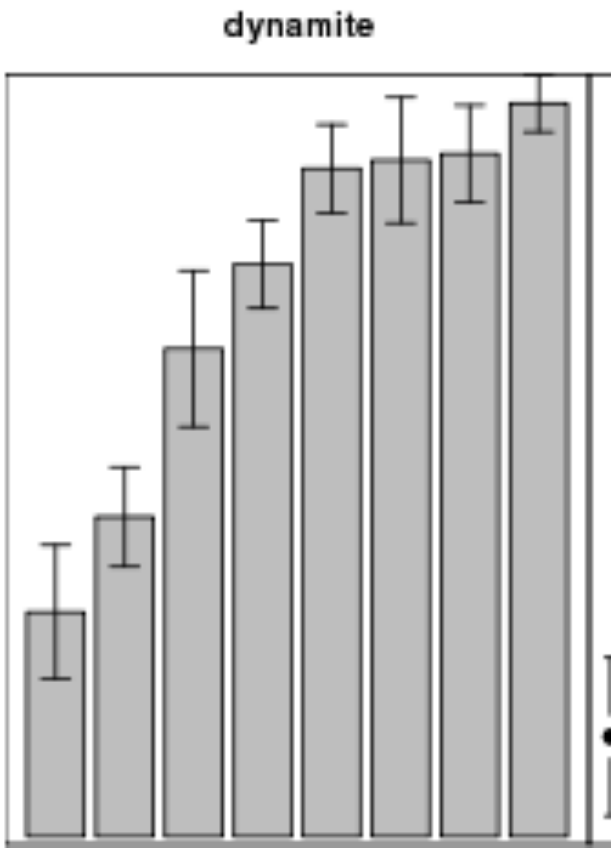
https://www.edwardtufte.com/tufte/

Books: https://www.amazon.com/Edward-R.-Tufte/e/B000APET3Y

Principles of Information Design

1. Tell the truth
2. Get to the point
3. Pick the right tool for the job
4. Highlight what is important
5. Keep it simple

Dynamite plots: an unmitigated evil?



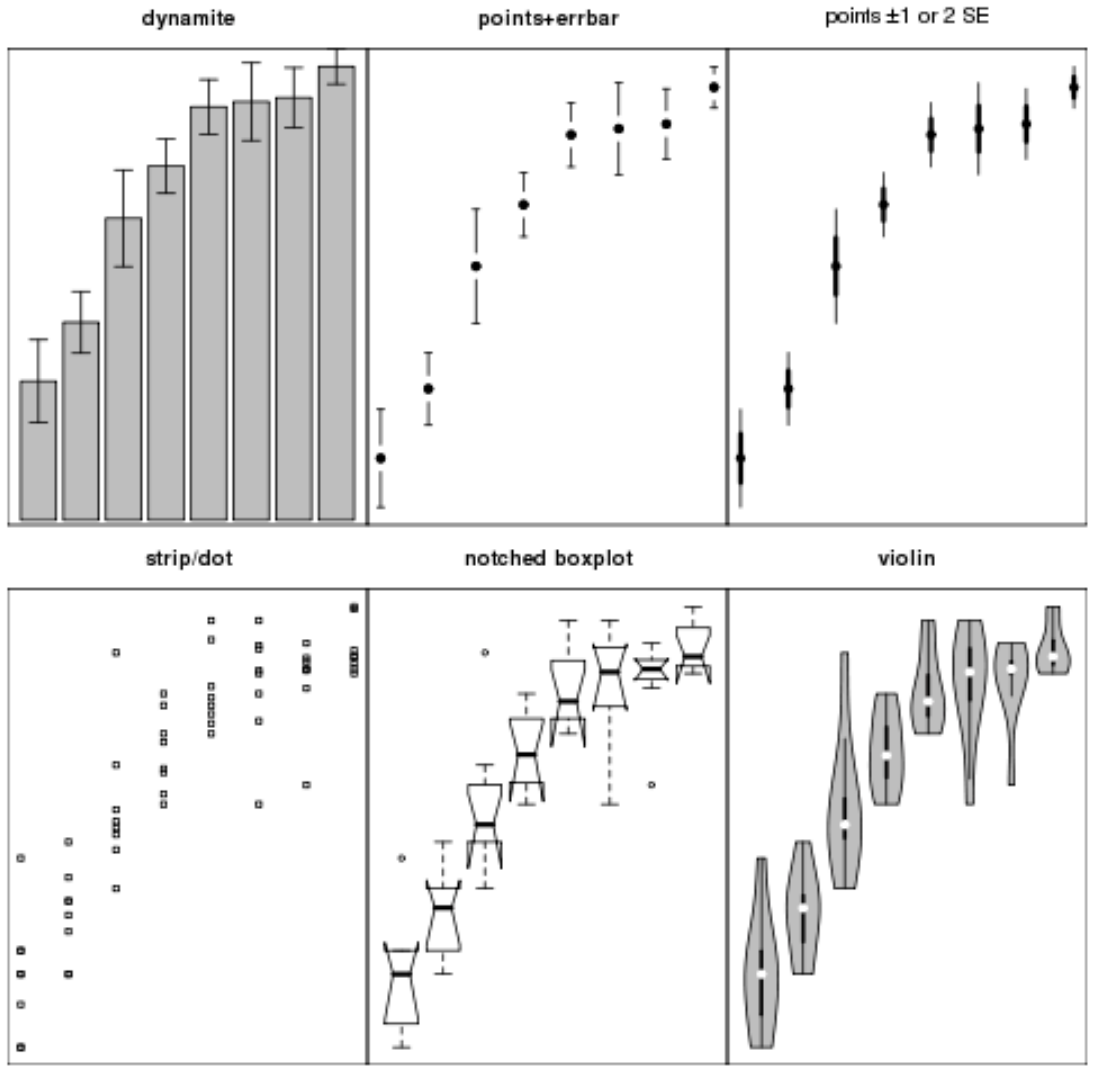
What is so bad about dynamite plots?

- They have low data-ink ratios
- They hide the raw data
- The (can) assume symmetric confidence bounds

What is good about dynamite plots?

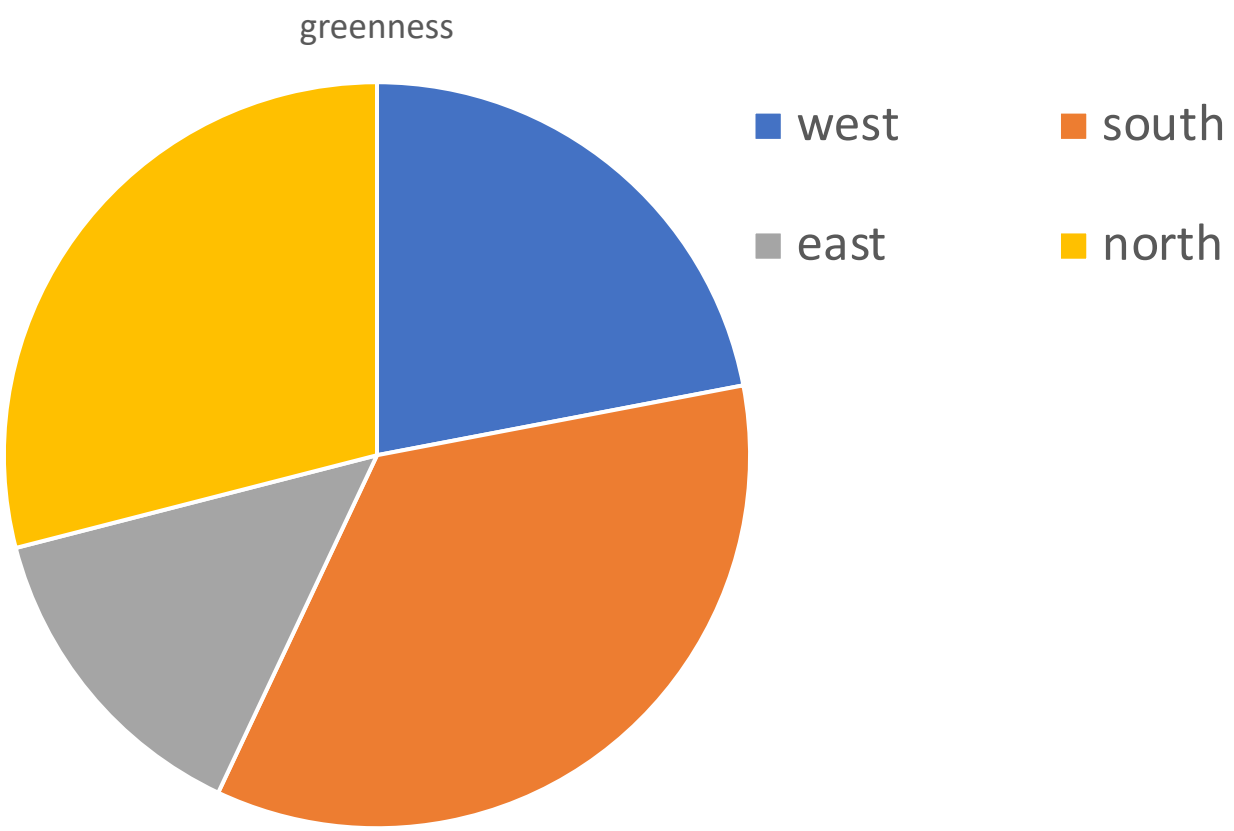
- they are comfortingly familiar to most people
- they anchor the data at zero (is this good?)
- they provide more area for displaying colors or gray scales than plots that use a single point for each value which may make it easier to distinguish groups of treatments in the data

Alternatives



Pies: better for eating

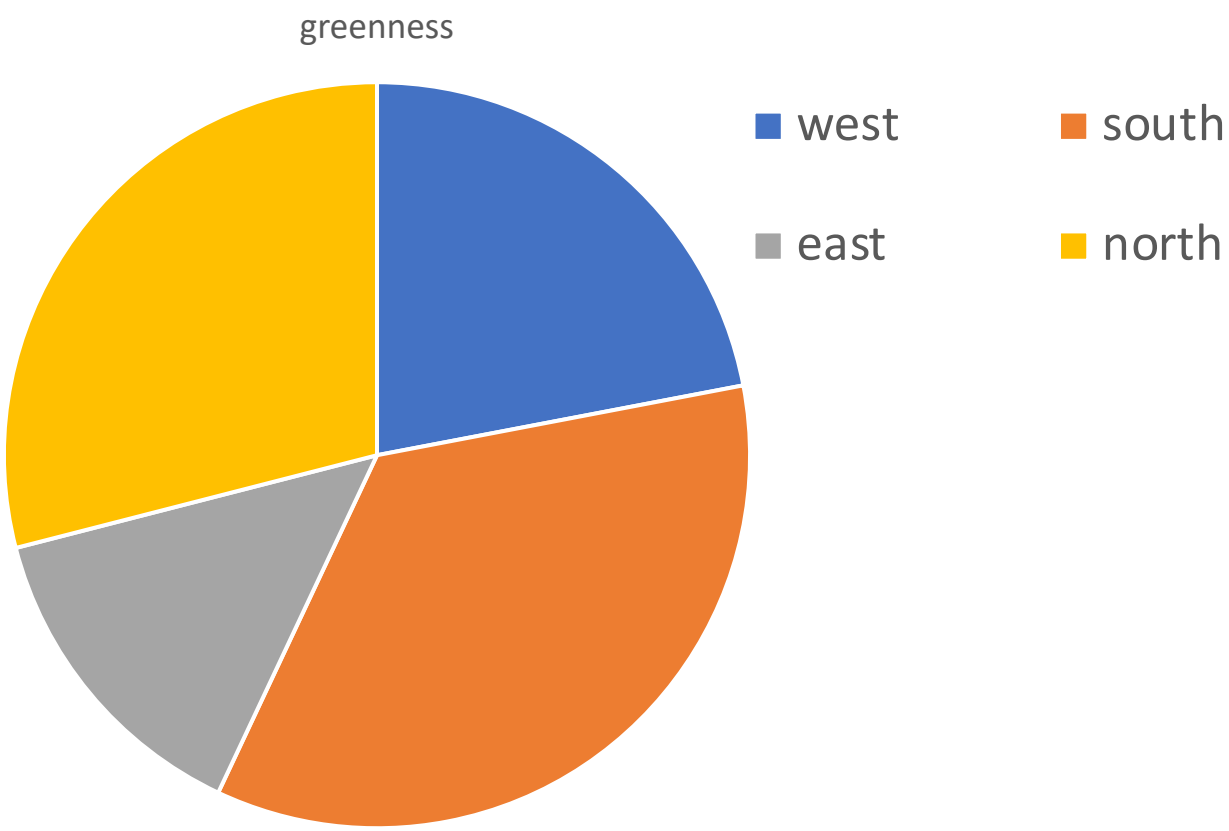
- Pie charts are not great for displaying data

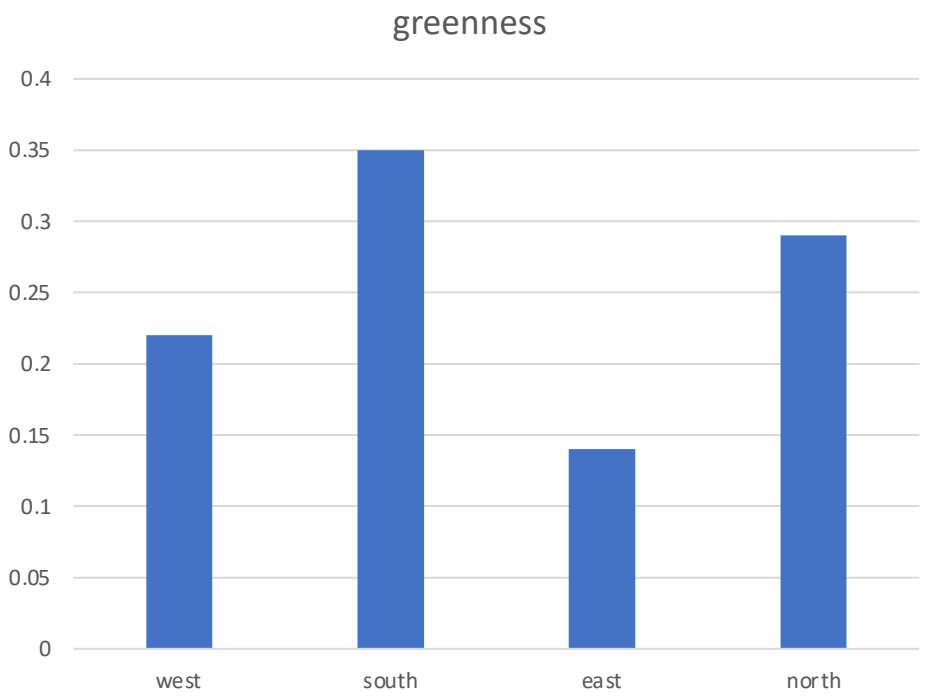


Compare the north and south – which is larger? How much larger?

Pies: better for eating

- Pie charts aren't great for displaying data

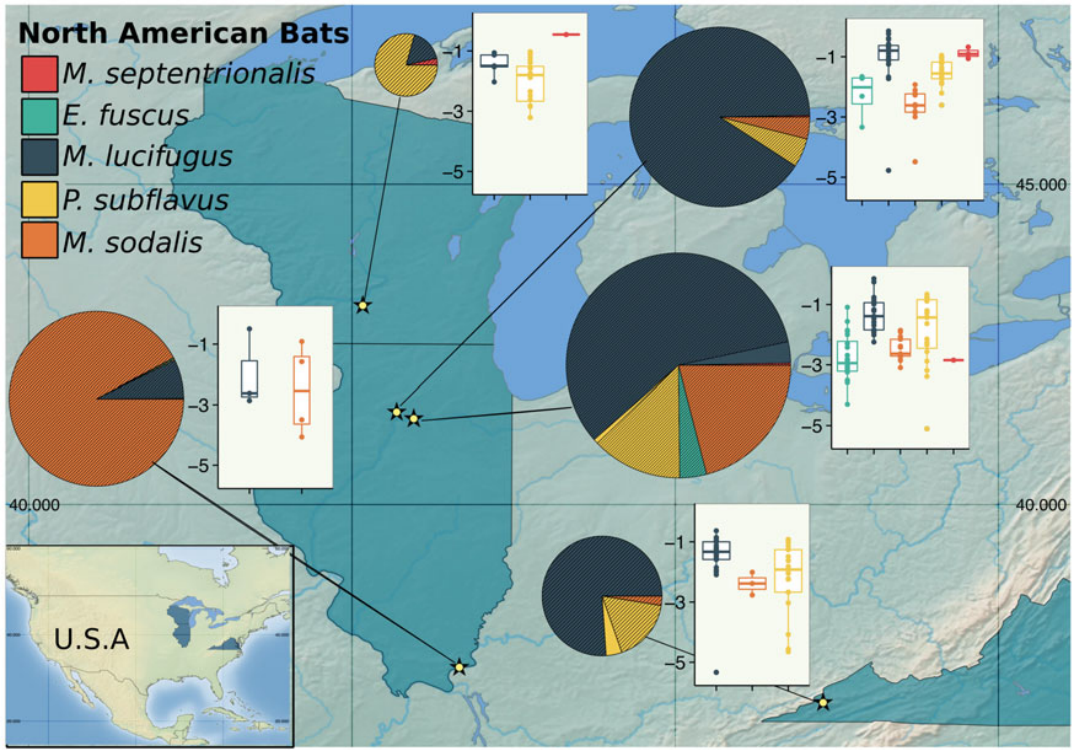




Pie chart caveats

- Pies can be somewhat useful for clearly displaying things that sum to 1 (e.g. 100%), and useful for eyeballing ½, ¼ etc.
- They can sometimes be helpful for providing descriptive detail that isn't critical for understanding of a figure, but you want to provide a reader that cares (e.g. data rich figs, fig limits).

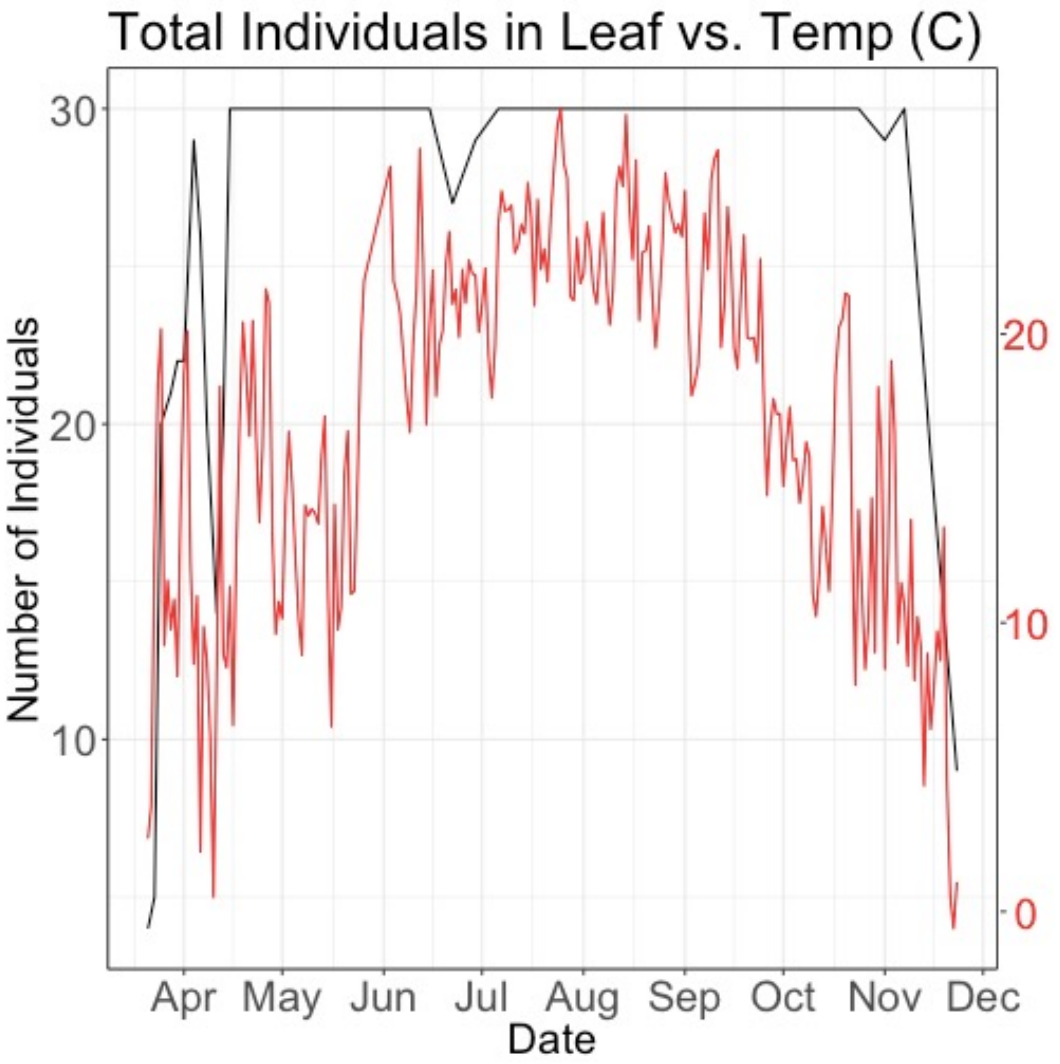
(a)



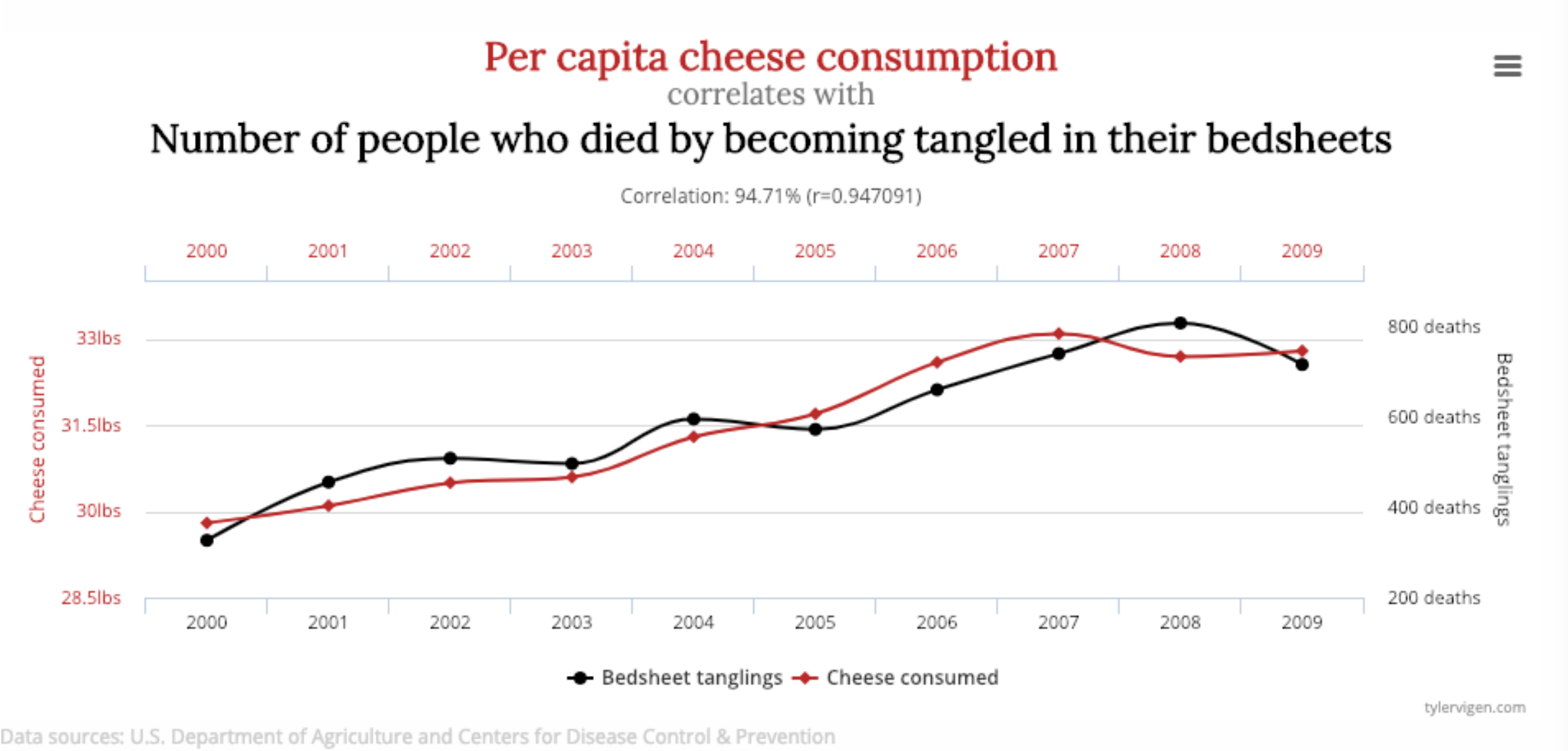
Dual axes graphs

Issues:

1) Scales can have a big impact on the viewer
2) They imply a causal relationship (very sloppy in time series)
3) They are often hideous



A nod to next week…(http://www.tylervigen.com/spurious-correlations)



Let the data speak vs storytelling

- Scientists feel differently about the relative value of each of these
- What do you see as the advantages or disadvantages of each?

The combination is critical

Nothing is more irritating to scientists than not being able to figure out what your data looks like. Only slightly less irritating is having to interpret your data because you didn’t do it for them.

YOUR job is to:

1) Tell your story.
   - put the output of statistics on your graphs
   - plot the important relationships
2) give the raw data to people so they can look at it themselves
   - don’t just plot the statistical output – overlay the points!
   - sometimes it isn’t practical to put all the raw data in a paper, but make tables, figs, etc. for online sources to show this to readers that want it

A few additional points of advice

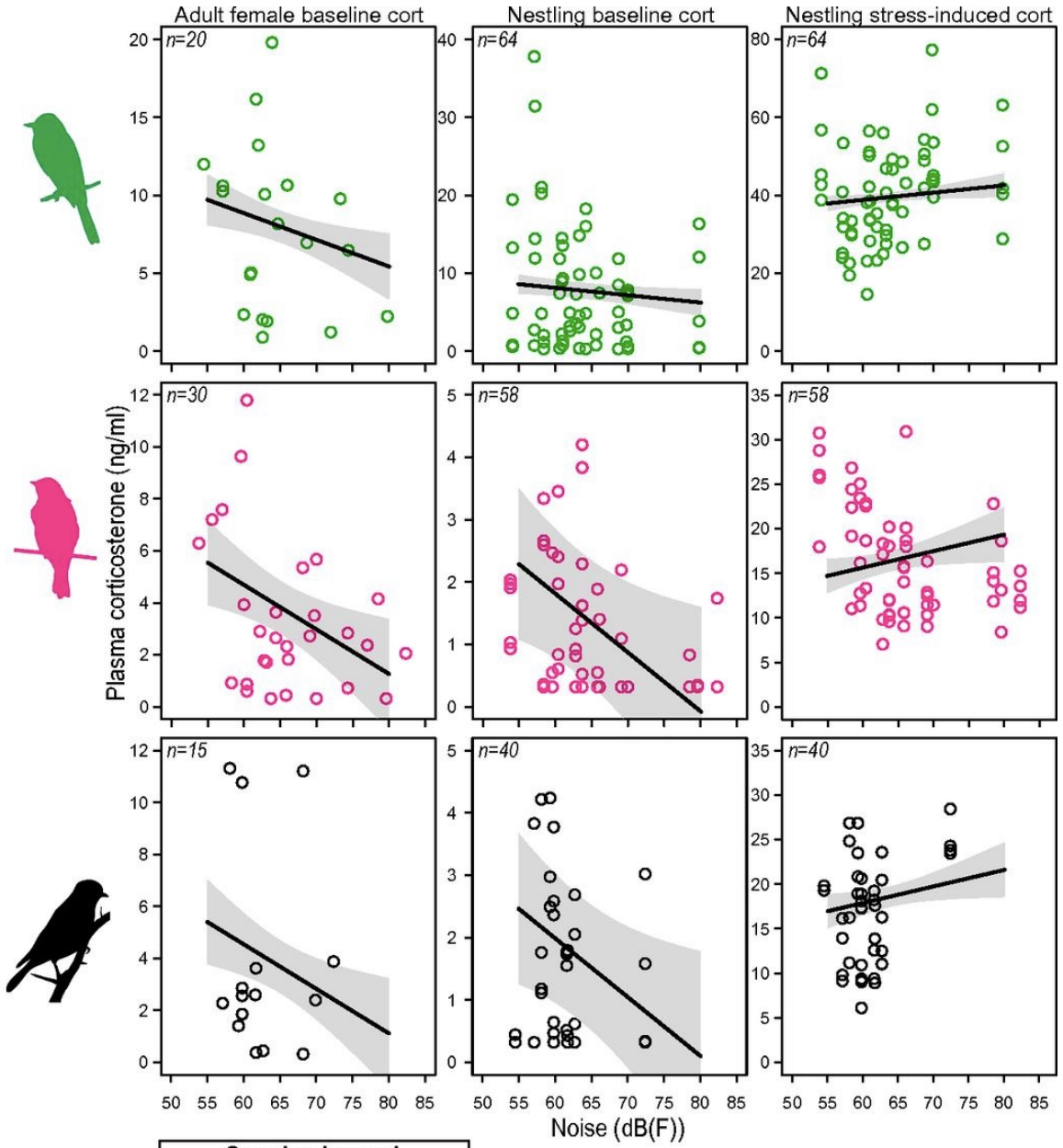
1) Consider your audience.

- Powerpoint vs print; novices vs field-specific, etc.

2) Don’t lie or censor data without good reason.

- Use consistent scales that represent your data, and don’t magnify the effect

3) Create figures that are compelling and tell a narrative.

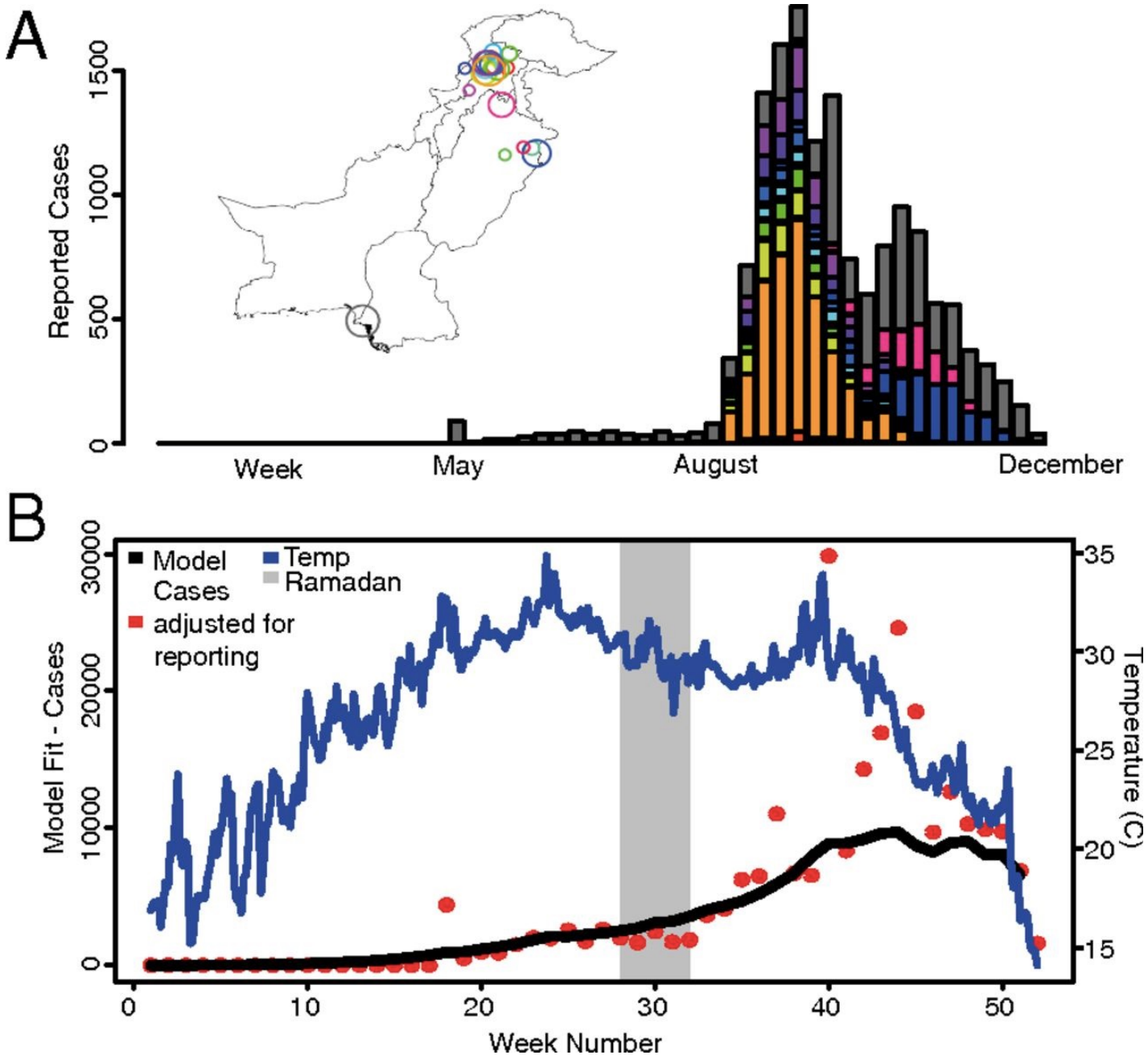


The graph shows the relationship between cort and noise in all three species and two life stages. Baseline cort levels for both adults and nestlings were negatively affected by noise (*Left* and *Center Columns*, respectively), while acute stress-induced cort levels in nestlings were positively affected by noise (*Right Column*). For each model, the effect of noise was consistent across species, irrespective of species-specific cort levels. The gray-shaded areas indicate the 85% CIs for each estimated effect.

Species legend

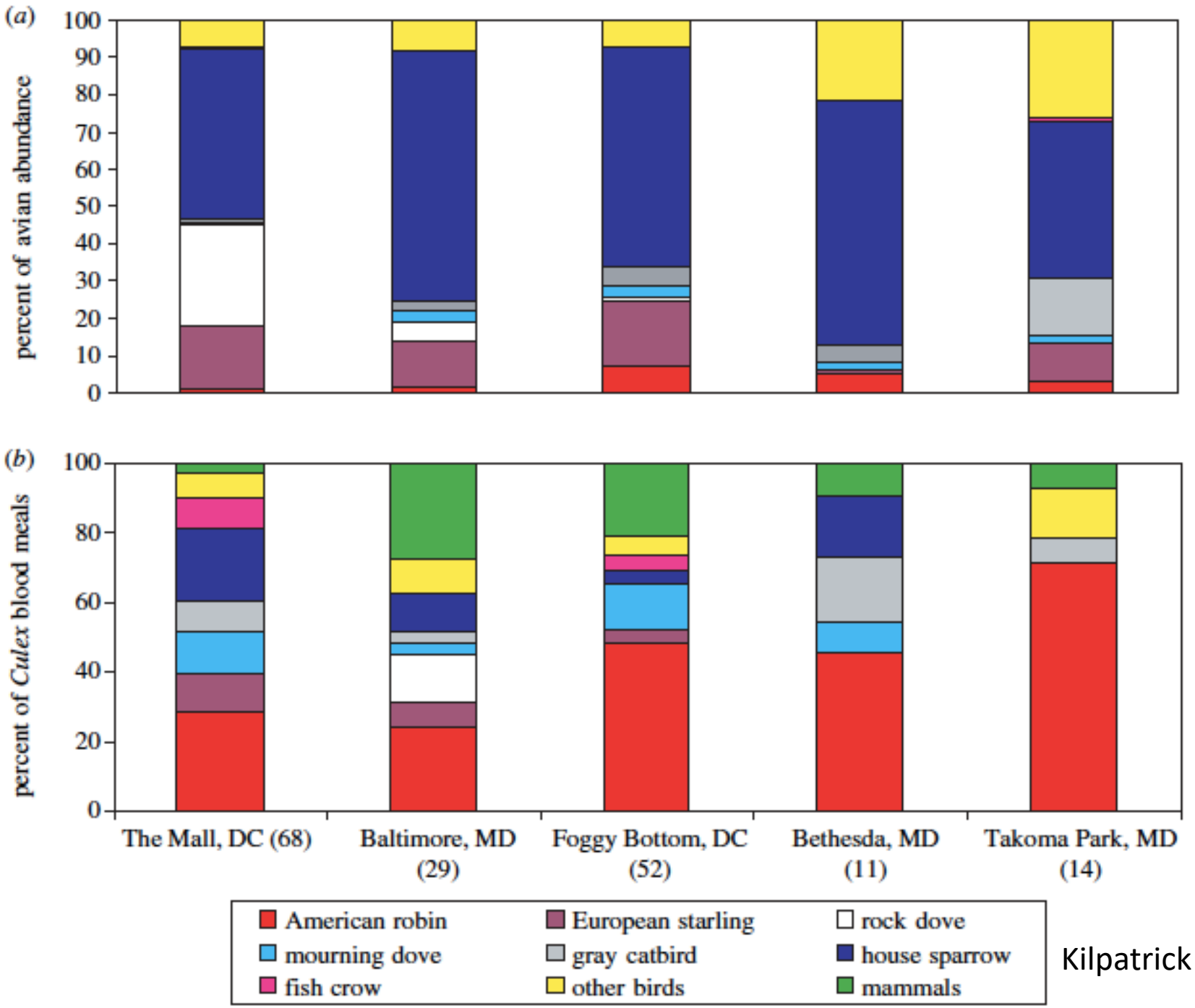
- Ash-throated flycatcher
- Western bluebird
- Mountain bluebird

Kleist 2018 PNAS



Dengue epidemiology in Pakistan in 2013. (*A*) The location of dengue cases throughout Pakistan and the number of cases per week by tehsil. Tehsils that reported at least 15 cases are shown on the map with corresponding color shown in the time series. The majority of cases were reported in Karachi (gray), Lahore (blue), and Mingora (orange). The dengue season in the entire country lasted 35 wk, with the first reported case in Karachi during week 18 (end of April). (*B*) The reported cases (red), temperature (blue), and model fit (black) for Karachi are shown. Using the case and temperature data, the human and vector population dynamics were modeled (***Materials and Methods***).

Wesolowski 2015 PNAS



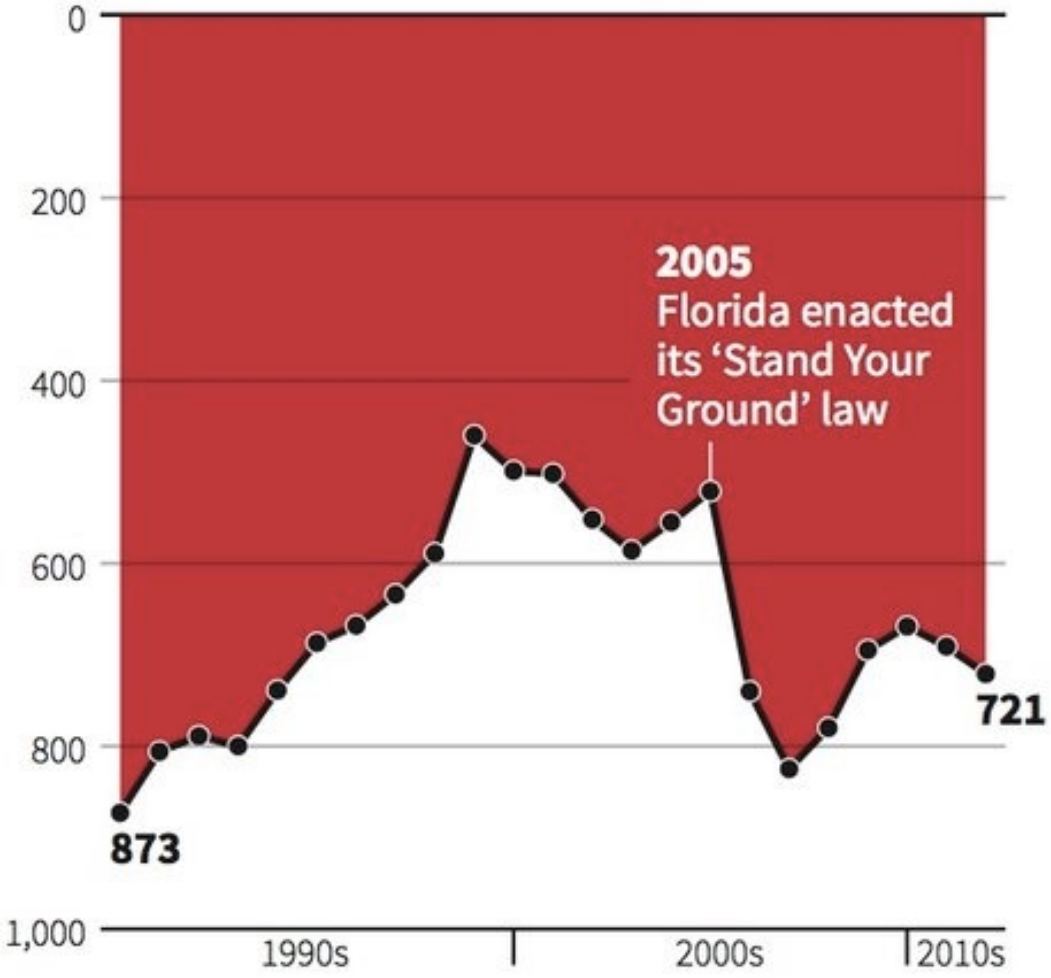
Kilpatrick 2006 Proc. Roy. Soc.

Animation as a tool for graph exploration

- https://nextstrain.org/ncov/europe?dmin=2020-07-29

Gun deaths in Florida

Number of murders committed using firearms

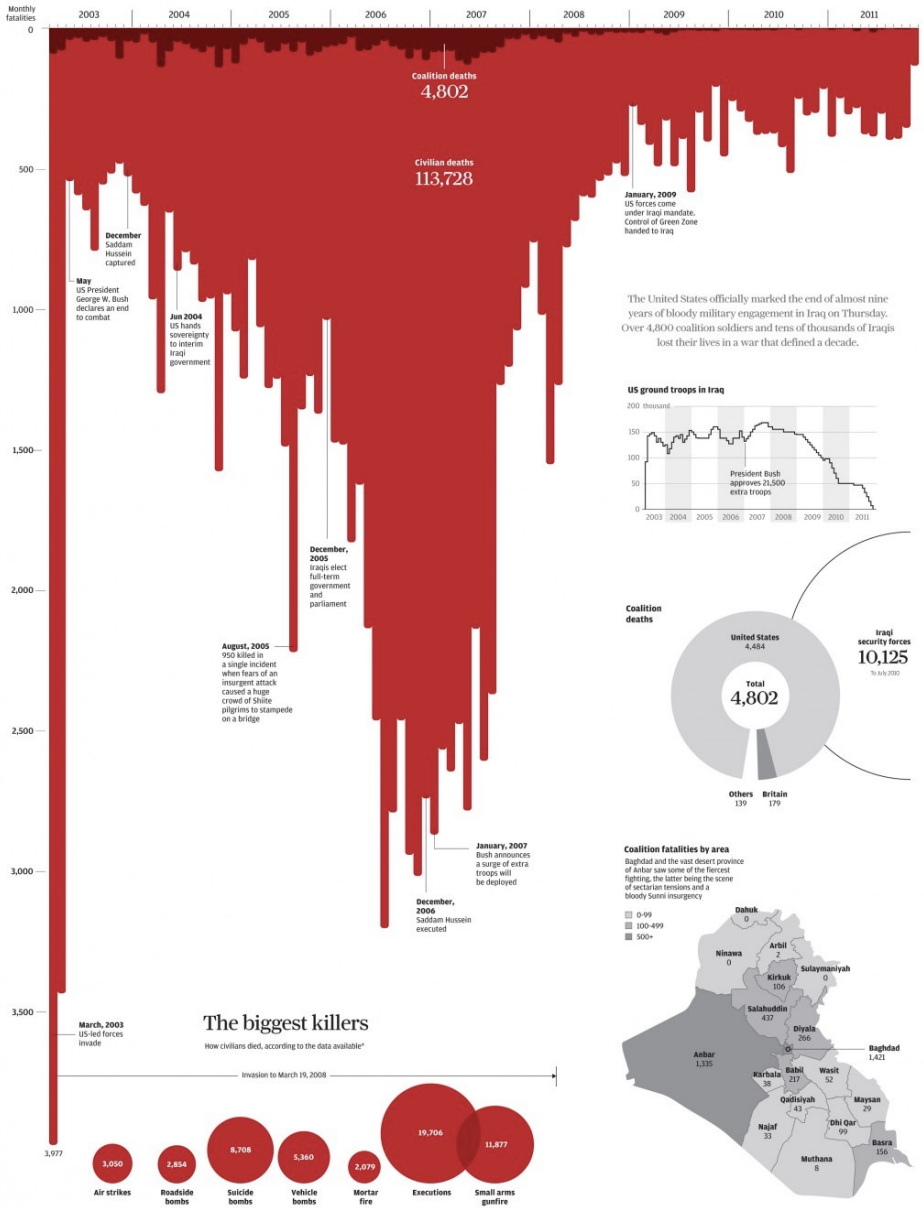


Source: Florida Department of Law Enforcement

C. Chan 16/02/2014

REUTERS

Iraq's bloody toll



The United States officially marked the end of almost nine years of bloody military engagement in Iraq on Thursday. Over 4,800 coalition soldiers and tens of thousands of Iraqis lost their lives in a war that defined a decade.

US ground troops in Iraq

The biggest killers

How civilians died, according to the data available*

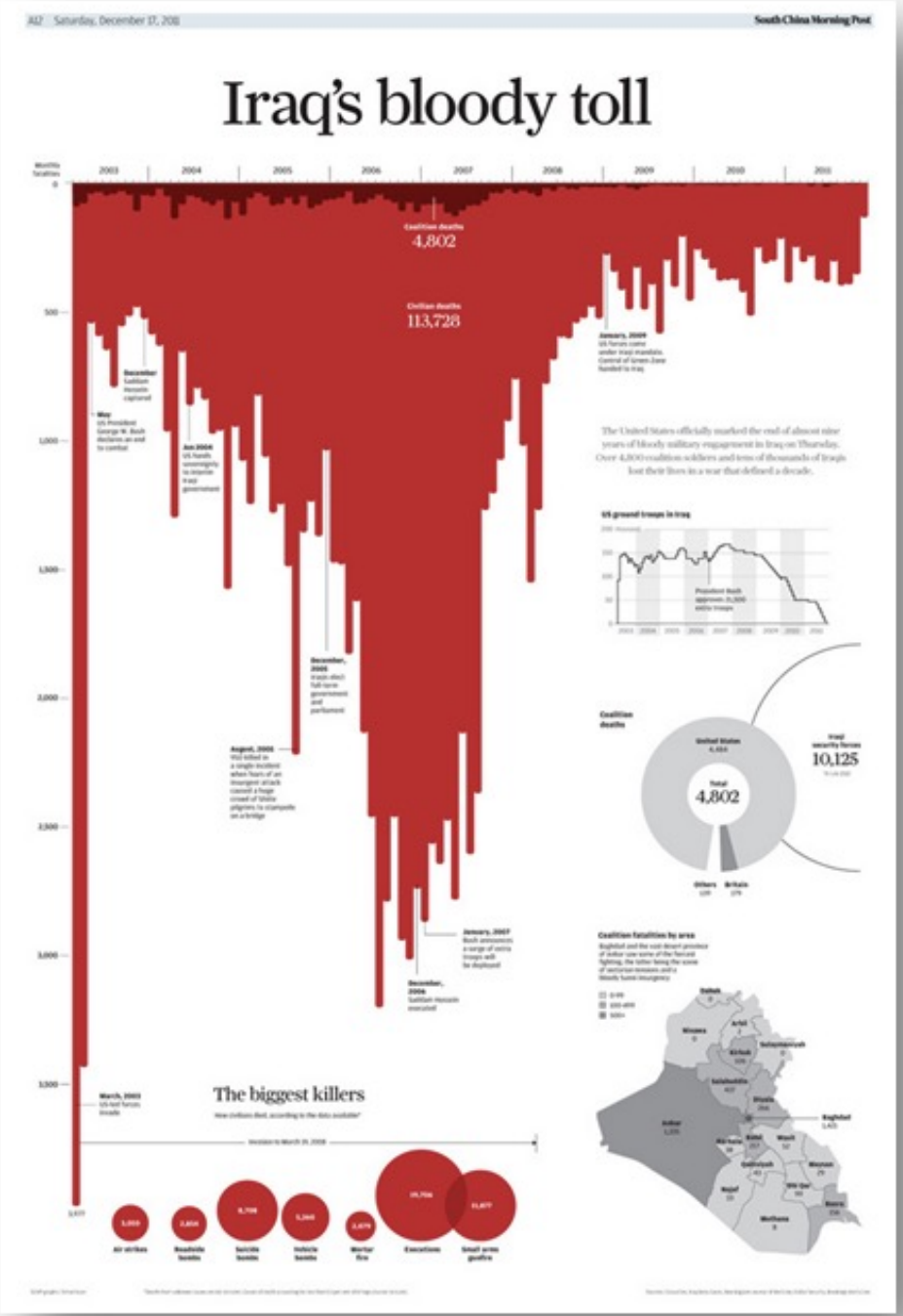
Coalition fatalities by area

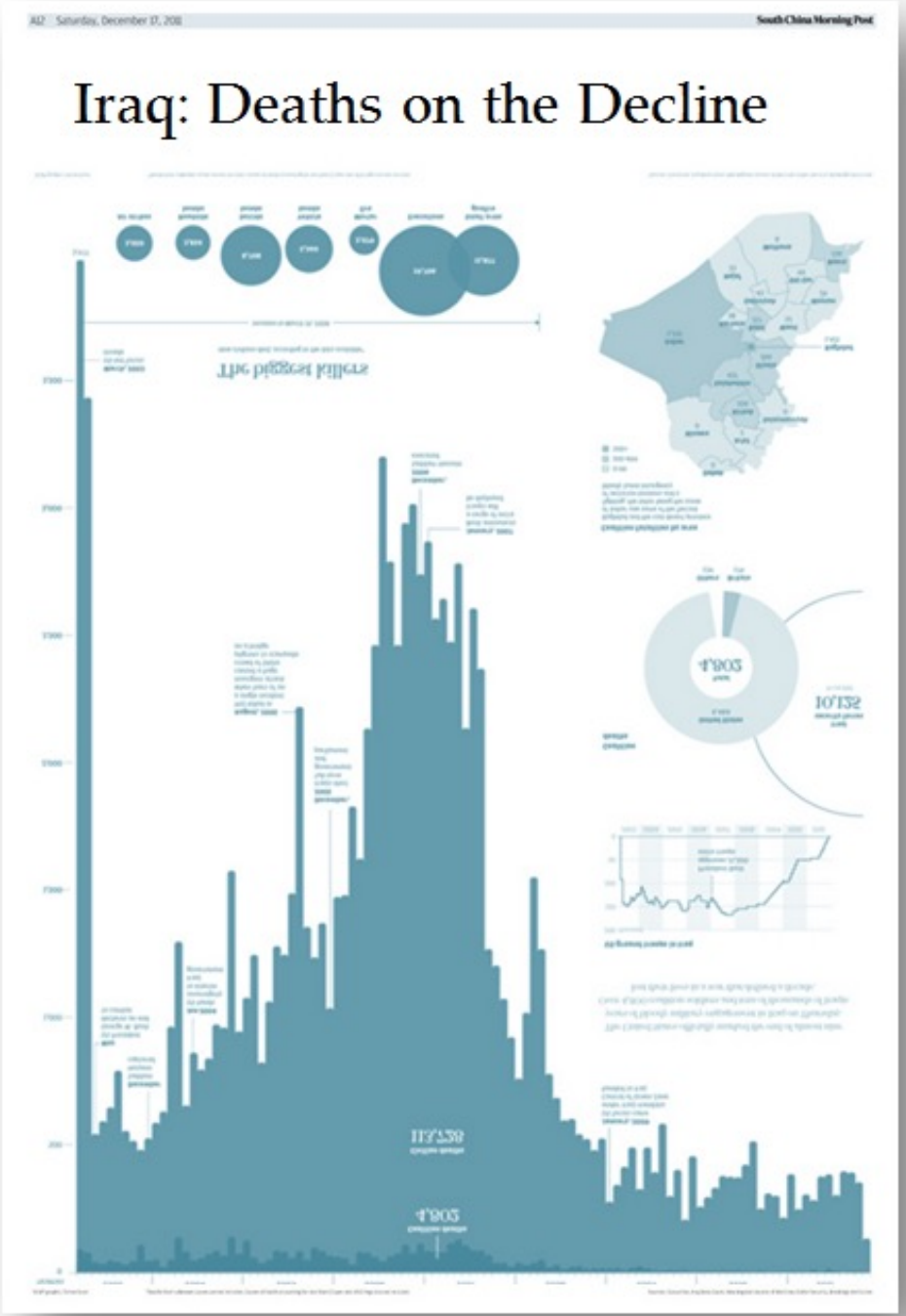
Baghdad and the vast desert province of Anbar saw some of the fiercest fighting, the latter being the scene of sectarian tensions and a bloody Sunni insurgency

SCMP graphic: Simon Scarr

*Deaths from unknown causes are not included. Causes of death accounting for less than 0.5 per cent of killings also not included.

Sources: iCasualties, Iraq Body Count, New England Journal of Medicine, Global Security, Brookings Institution


A12 Saturday, December 17, 2011
South China Morning Post
Iraq's bloody toll
Coalition deaths
4,802
Civilian deaths
113,728
The United States officially marked the end of almost nine years of bloody military engagement in Iraq on Thursday. Over 4,800 coalition soldiers and tens of thousands of Iraqis lost their lives in a war that defined a decade.
US ground troops in Iraq
Coalition deaths
Total
4,802
Iraqi security forces
10,125
The biggest killers
Coalition fatalities by area


A12 Saturday, December 17, 2011
South China Morning Post
Iraq: Deaths on the Decline

ggplot2

- Best website for help: http://ggplot2.tidyverse.org/reference/
- **aesthetic mappings**: between variables in the data frame and **aesthetics**, or graphical attributes (x position, y position, size, colour ...)
- first two show up as (e.g.) ggplot(my_data, aes(x=age,y=rootgrowth,colour=phosphate))
- **geoms**:
  - simple: geom_point, geom_line:
  - For great reference: www.r-graph-gallery.com

  - SWITCH TO CONSOLE